Table D1. Temporal structures estimated as random parameters binary logit models with heterogeneity in means and variances, by approach and selection stage, Thailand, 2016–2023.

Temporal approach	Structure selected in	Candidate[a]	Penalty λ	Parameters at selection[b]	Confidence of rejection at selection[b]	Parameters after rebuilding and pruning	LL(β)	AIC	BIC	LR test against the fully unconstrained model, χ^2 (df)	Confidence of rejection	Outcome
S1: single-motorcycle crashes (two-year periods); N = 7,325												
Partially constrained	Fixed-parameter logit	–	–	73	0.023	52	–4,526.94	9,157.9	**9,516.6**	33.7 (40)	0.251	Reported (Table 3)
Partially constrained	Random parameters model	–	–	77	0.000	54	–4,520.58	9,149.2	9,521.7	21.0 (38)	0.011	Not rejected
Adaptive fused	Fixed-parameter logit	–	8.85×10^{-4}	59	0.875	45	–4,534.14	9,158.3	**9,468.7**	48.1 (47)	0.571	Reported (Table 3)
Adaptive fused	Random parameters model	1	1.01×10^{-3}	59	0.893	39	–4,553.86	9,185.7	9,454.8	87.5 (53)	0.998	Rejected → next candidate
Adaptive fused	Random parameters model	2	7.60×10^{-4}	64	0.415	39	–4,560.40	9,198.8	9,467.9	100.6 (53)	1.000	Rejected → next candidate
Adaptive fused	Random parameters model	3	5.70×10^{-4}	67	0.147	43	–4,552.46	9,190.9	9,487.6	84.7 (49)	0.999	Rejected → next candidate
Adaptive fused	Random parameters model	4	4.27×10^{-4}	73	0.001	49	–4,527.26	9,152.5	9,490.6	34.3 (43)	0.175	Not rejected
S2: motorcycle and another vehicle type; N = 21,371												
Partially constrained	Fixed-parameter logit	–	–	109	0.000	85	–11,979.04	24,128.1	**24,805.5**	–62.3 (154)	0.000	Reported (Table 3)
Partially constrained	Random parameters model	–	–	116	0.000	86	–11,992.49	24,157.0	24,842.4	–35.4 (153)	0.000	Not rejected
Adaptive fused	Fixed-parameter logit	–	4.17×10^{-4}	66	0.887	59	–12,088.86	24,295.7	24,765.9	157.3 (180)	0.112	Not rejected
Adaptive fused	Random parameters model	1	5.29×10^{-4}	68	0.847	56	–12,052.03	24,216.1	**24,662.4**	83.6 (183)	0.000	Reported (Table 3)
S3: single vehicle, no motorcycle; N = 73,731												
Partially constrained	Fixed-parameter logit	–	–	103	0.000	83	–23,590.45	47,346.9	**48,111.2**	69.0 (155)	0.000	Reported (Table 3)
Partially constrained	Random parameters model	–	–	116	0.000	86	–23,597.42	47,366.8	48,158.7	82.9 (152)	0.000	Not rejected
Adaptive fused	Fixed-parameter logit	–	1.25×10^{-4}	69	0.889	65	–23,646.87	47,423.7	**48,022.3**	181.8 (173)	0.692	Reported (Table 3)
Adaptive fused	Random parameters model	1	1.05×10^{-4}	78	0.868	63	–23,662.90	47,451.8	48,031.9	213.9 (175)	0.976	Rejected → next candidate
Adaptive fused	Random parameters model	2	7.91×10^{-5}	90	0.063	64	–23,698.91	47,525.8	48,115.1	285.9 (174)	1.000	Rejected → next candidate
Adaptive fused	Random parameters model	3	5.93×10^{-5}	104	0.001	82	–23,618.52	47,401.0	48,156.1	125.1 (156)	0.033	Not rejected
S4: multi-vehicle, no motorcycle; N = 33,646												
Partially constrained	Fixed-parameter logit	–	–	99	0.000	88	–12,586.54	25,349.1	26,090.4	89.6 (127)	0.005	Not rejected
Partially constrained	Random parameters model	–	–	103	0.290	81	–12,598.28	25,358.6	**26,040.9**	113.1 (134)	0.095	Reported (Table 3)

Temporal approach	Structure selected in	Candidate[a]	Penalty λ	Parameters at selection[b]	Confidence of rejection at selection[b]	Parameters after rebuilding and pruning	LL(β)	AIC	BIC	LR test against the fully unconstrained model, χ^2 (df)	Confidence of rejection	Outcome
Adaptive fused	Fixed-parameter logit	–	2.53×10^{-4}	68	0.881	65	–12,643.53	25,417.1	25,964.6	203.6 (150)	0.998	Rejected
Adaptive fused	Random parameters model	1	1.31×10^{-4}	95	0.890	72	–12,617.43	25,378.9	**25,985.4**	151.4 (143)	0.701	Reported (Table 3)

Notes. All models: 2,000 Halton draws; random parameters and heterogeneity terms searched with 100 draws and confirmed with 2,000 draws; non-significant parameters removed (90% level). The BIC of the reported model of each approach is in bold. Partially constrained: adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level, in the fixed-parameter logit or inside the random parameters model (random parameters and heterogeneity terms of the fixed-parameter-stage model held fixed during merging). Adaptive fused: sparsest penalty λ whose structure is not rejected at the 90% level ($LR < \chi^2_{0.90}(df)$), in the fixed-parameter logit or inside the random parameters model. Each structure is then rebuilt (search for random parameters and heterogeneity terms) and pruned, and tested against the pruned fully unconstrained (separate period) random parameters models. For each approach, the stage with the lower BIC among models not rejected at the 90% level is reported in Table 3.

[a] Structures selected inside the random parameters model by the adaptive fused lasso are tried in order of sparsity (decreasing λ); a candidate rejected after rebuilding and pruning is replaced by the next one.

[b] Before rebuilding and pruning. Fixed-parameter logit: parameters of the fixed-parameter model and confidence of rejection against the separate-period logits (Table 2). Random parameters model: parameters of the selected structure with the random parameters and heterogeneity terms held fixed, and confidence of rejection against the separate period random parameters models before pruning.